

ENG507

Consonantal Gestures

1. **Consonantal Gestures**
Movements of speech organs (like the tongue, lips, or throat) to produce consonant sounds.
 2. **Articulatory Targets**
The specific positions or configurations that the speech organs aim for when producing sounds.
 3. **Types of Articulatory Gestures**
 - **Stops:** Sounds made by completely blocking airflow (e.g., *p, t*).
 - **Nasals:** Sounds made with airflow through the nose (e.g., *m, n*).
 - **Fricatives:** Sounds made by narrowing the airway and creating friction (e.g., *f, s*).
 - **Trills, Taps, and Flaps:** Rapid movements of the tongue or other articulators (e.g., rolling *r*).
 - **Laterals:** Sounds made by allowing airflow along the sides of the tongue (e.g., *l*).
 4. **Summary of Articulatory Gestures**
A summary of how different speech sounds are formed using specific movements of speech organs.
-

Acoustic Phonetics

1. **Acoustic Phonetics**
The study of the physical properties of speech sounds, like pitch, loudness, and duration.
2. **Acoustic Analysis**
Using tools (like spectrograms) to study the sound waves of speech.
3. **Acoustic Analysis of Vowels**
Examining vowel sounds based on their frequencies and patterns.
4. **Source-Filter Theory of Speech Production**
Explains how sound is created (source) and shaped (filter) in the vocal tract to produce speech.
5. **Tube Models**
A way to understand the vocal tract as a series of tubes that modify sound.
6. **Perturbation Theory**
Describes how changes in the shape of the vocal tract affect the sound frequencies.
7. **Acoustics of Consonants**
The study of how consonant sounds differ in their acoustic properties.

8. Spectrograms

Visual representations of sound, showing frequency, intensity, and time.

Vowels

1. Cardinal Vowels

A set of standard vowel sounds used as reference points.

2. Secondary Cardinal Vowels

Variations of primary cardinal vowels with different tongue positions or lip shapes.

3. Vowels in BBC English

The specific vowel sounds used in British English.

4. Advanced Tongue Root (ATR)

A feature of some languages where the tongue is pulled forward during vowel production.

5. Rhotacized Vowels

Vowels with a distinct *r*-like quality.

6. Nasalization in Vowels

Vowel sounds produced with airflow through the nose.

Semivowels and Secondary Articulatory Gestures

1. Semivowels

Sounds like vowels but function as consonants (e.g., *y* in *yes*).

2. Secondary Articulatory Gestures (SAG)

Additional movements of the speech organs that modify sounds (e.g., lip rounding or tongue height).

Suprasegmental Features

1. Introduction to Suprasegmental (SS) Features

Features of speech that go beyond individual sounds, like stress, pitch, and intonation.

2. Syllable

A unit of sound in a word, typically containing a vowel sound (e.g., *cat* has one syllable).

3. Stress

Emphasis placed on certain syllables or words.

4. Stress-Timed Languages

Languages where stressed syllables occur at regular intervals (e.g., English).

5. Syllable-Timed Languages

Languages where each syllable takes the same amount of time (e.g., French).

6. **Pitch**
The highness or lowness of a sound.
7. **Tone and Tonal Languages**
Languages where pitch changes the meaning of a word (e.g., Chinese).
8. **Intonation**
The rise and fall of pitch in speech to convey meaning or emotion.
9. **Functions of Intonation**
 - Showing emotions.
 - Indicating questions or statements.
 - Highlighting important words.

Linguistic Phonetics

1. **Phonetics of the Community and of the Individual**
 - **Community:** Studies how sounds are used by a group of people who share a language or dialect.
 - **Individual:** Focuses on the unique way a person pronounces sounds.
2. **The IPA (International Phonetic Alphabet)**
A system of symbols used to represent all the sounds of human speech. It helps describe pronunciation precisely.
3. **Feature Hierarchy**
A way of organizing speech sounds based on shared characteristics, like voicing (e.g., *b* is voiced, *p* is voiceless).
4. **Problems with Linguistic Explanations**
Challenges in fully explaining how people produce and perceive sounds because of individual variation and complexity.
5. **Controlling Articulatory Movements**
How the brain and muscles coordinate to produce precise speech sounds.
6. **Memory for Speech**
The ability to remember and reproduce sounds, words, and speech patterns.
7. **The Balance Between Phonetic Forces**
The way different factors (like ease of pronunciation and clarity) shape how speech sounds are formed.

Phonotactics and Syllabic Templates

1. **Syllabic Templates and Syllabification**
 - **Templates:** Rules about how sounds combine to form syllables (e.g., a syllable often has a vowel in the middle).
 - **Syllabification:** Breaking words into syllables (e.g., *win-dow* has two syllables).
2. **Phonotactics**
Rules about which sounds can appear together in a language (e.g., *ng* is common in English at the end of words like *sing* but not at the beginning).

Using PRAAT

1. Introduction to PRAAT

A software tool used to study and analyze speech sounds visually and acoustically.

2. Exploring PRAAT

- **Recording and Displaying:** Record speech and view its sound waves.
- **Segmenting and Labeling:** Divide the sound file into parts and label them.
- **Exporting Visual Displays:** Save spectrograms or waveforms for use in documents.

3. The Source-Filter Model of Speech

Explains how the vocal cords produce sound (source), and how the vocal tract shapes it (filter).

4. Measuring Speech Features

- **Fundamental Frequency:** The basic pitch of the voice.
- **Harmonics:** Higher frequencies that add richness to sound.
- **Formants:** Resonant frequencies shaped by the vocal tract, important for distinguishing vowels.

5. Vowel Properties and Charts

Plotting vowels on a chart based on their formants to study their acoustic properties.

6. Other Measurements

- **Sonorants, Nasal Formants, and Glides:** Characteristics of smoother sounds like *m*, *n*, or *y*.
- **Stop Voicing on Spectrograph:** Shows whether a sound like *b* or *p* is voiced.
- **Voice Onset Time (VOT):** Measures the time between releasing a stop sound (like *p*) and starting to voice it.

Further Areas of Study in Phonetics and Phonology (P&P)

1. Current Trends in P&P Research

- Investigating sound patterns in new languages or dialects.
- Exploring how technology can improve pronunciation teaching.

2. Distinctive Features

Characteristics that differentiate one sound from another (e.g., voiced vs. voiceless).

3. Experimental Phonetics

Using experiments to understand how speech sounds are produced and perceived.

4. The Study of Variety

Examining how pronunciation changes between accents or regions.

The Pedagogy of Phonetics and Phonology

1. **Relationship Between ELT (English Language Teaching) and P&P**
Phonetics and phonology help teachers improve learners' pronunciation and listening skills.
2. **Developing Relevant Material**
Creating exercises and tools to teach sounds and pronunciation effectively.
3. **Conducting Classroom Research for ELT**
Studying how students learn pronunciation to find better teaching methods.
4. **Making Research Accessible to Teachers**
Sharing findings in simple ways so teachers can apply them in their classrooms.
5. **Facilitating Action Research**
Encouraging teachers to explore their own classrooms to improve pronunciation teaching.

